

SEQUENTIAL SYSTEMS

Phase I: The Genesis of Memory

A Comprehensive Master Manual

Topics Covered:

Combinational vs. Sequential Logic

Feedback & Latch Physics

Clocking & Edge-Triggering

Master-Slave Architecture

The Four Core Flip-Flops

Excitation & State Diagrams

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Chapter 1

Foundations of Sequential Logic

1.1 The Limitation of Combinational Logic

In the study of digital electronics, the initial focus is always placed upon Combinational Logic. These circuits include Adders, Subtractors, Multiplexers, Demultiplexers, Encoders, and Decoders. The defining characteristic of a combinational circuit is that its output is strictly and exclusively a mathematical function of its *present* inputs.

Mathematically, if we denote the input vector at time t as $X(t)$, the output vector $Y(t)$ is given by:

$$Y(t) = f(X(t)) \quad (1.1)$$

This architecture possesses a critical flaw when attempting to build a computing machine: **It has zero memory.** If the input signals are removed or lose power, the output immediately ceases to exist. A processor cannot perform sequential algorithms if it cannot remember the result of the previous calculation.

Combinational circuits are essentially "feed-forward" networks. Data flows linearly from the input pins, through the logic gates, and out to the destination pins. Because there is no mechanism to trap or sustain a signal, these circuits live entirely in the

present moment.

1.2 The Introduction of Sequential Logic

To construct a computer, we require a circuit that can retain a logical state (a '1' or a '0') indefinitely, long after the triggering input has been removed. This requires the introduction of **Sequential Logic**.

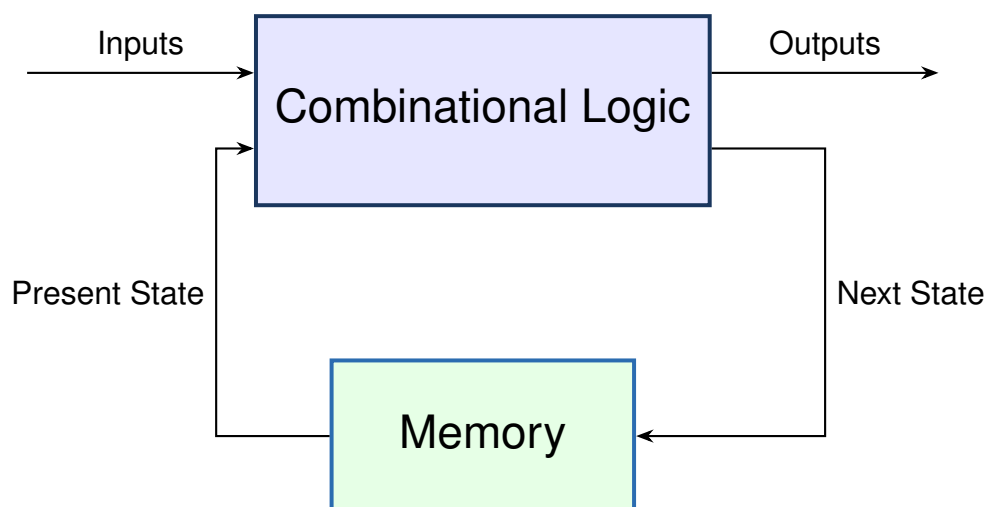
A Sequential Logic circuit is formally defined as a combinational logic circuit augmented with **Memory Elements**. Furthermore, it introduces the critical concept of *Feedback*.

1.2.1 The Feedback Loop

The only physical method to create memory without mechanical switches is to route the output of a logic gate directly back into its own input. By feeding the output back into the system, the signal becomes "trapped" in a stable loop, sustaining itself.

Mathematically, the output Y at time $t + 1$ depends on the input at time t *and* the output state at time t :

$$Y(t + 1) = f(X(t), Y(t)) \quad (1.2)$$



1.3 Comparative Analysis

To solidify this foundational concept, we establish the absolute differences between the two logic families.

Parameter	Combinational Logic	Sequential Logic
Output Dependency	Depends only on present inputs.	Depends on present inputs and past states.
Memory	Entirely memory-less.	Contains memory elements (Latches/FFs).
Feedback	No feedback path exists.	Feedback path is mandatory.
Clock Signal	Operates asynchronously.	Typically governed by a Clock (Synchronous).
Examples	Adders, MUX, Decoders.	Counters, Registers, FSMs.

Chapter 2

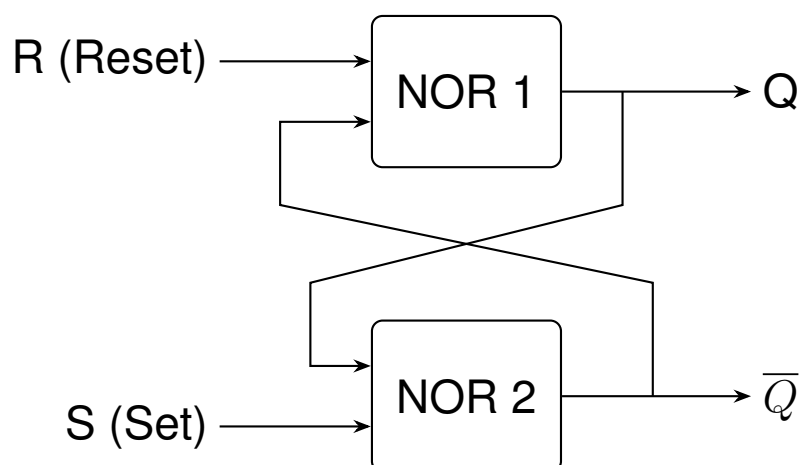
The Latches: Genesis of Memory

2.1 The SR Latch (Set-Reset)

The most fundamental, primitive memory element in digital electronics is the **SR Latch**. It is constructed by cross-coupling two NOR gates (or two NAND gates).

The term "cross-coupled" means the output of the first NOR gate is wired directly into one of the inputs of the second NOR gate, and the output of the second is wired back into the first. This physical wiring is what creates the feedback loop capable of trapping a 1 or 0 indefinitely.

2.1.1 Logic Diagram of NOR-based SR Latch



2.2 Operational Mechanics of the SR Latch

The SR latch has two inputs (S and R) and two outputs (Q and $\overline{Q}$). By fundamental design, the outputs must *always* be logical complements of each other. If $Q=1$, $\overline{Q}$ must be 0. This complement rule is the absolute law of the memory element.

Let us analyze the four possible input conditions logically:

2.2.1 Condition 1: Set (S=1, R=0)

When the Set pin is pulsed HIGH, the logic of the cross-coupled NOR gates forces the primary output **Q to become 1**. Because Q becomes 1, the feedback loop forces the complementary output $\overline{Q}$ to become 0. The latch has effectively stored a logical '1'.

2.2.2 Condition 2: Reset (S=0, R=1)

When the Reset pin is pulsed HIGH, the logic is reversed. The primary output **Q is forced to 0**, and $\overline{Q}$ becomes 1. The latch has been cleared, storing a logical '0'.

2.2.3 Condition 3: Hold / Memory (S=0, R=0)

This is the most critical and magical state in computing. When both inputs are dropped to 0 (imagine you remove your fingers from the Set and Reset buttons), the cross-coupled feedback loops sustain the current voltages. If Q was 1, it stays 1. If Q was 0, it stays 0. **This state is the physical manifestation of digital memory.**

2.2.4 Condition 4: The Forbidden State (S=1, R=1)

If both Set and Reset are driven HIGH simultaneously, the NOR gate mathematical logic breaks down. Both Q and $\overline{Q}$ are forced to 0. This violates the sacred rule that they must be complements. Worse, if both inputs drop back to 0 at the exact same mi-

crosecond, the circuit enters a *metastable* state and will randomly collapse into either a 1 or a 0 unpredictably. Therefore, this state is strictly forbidden in hardware design.

2.3 Mathematical Modeling of the SR Latch

To utilize the SR Latch in engineering design, we cannot rely solely on textual descriptions. We must compress its physical behavior into rigorous mathematical tables.

2.3.1 The Characteristic Truth Table

This table predicts the next state (Q_{n+1}) based on the prevailing inputs and the present state (Q_n).

S	R	Q_n	Q_{n+1}	System Status
0	0	0	0	Hold (Remembers 0)
0	0	1	1	Hold (Remembers 1)
0	1	0	0	Reset
0	1	1	0	Reset
1	0	0	1	Set
1	0	1	1	Set
1	1	0	X	Forbidden / Invalid
1	1	1	X	Forbidden / Invalid

2.3.2 The Characteristic Equation

By mapping the valid states of Q_{n+1} onto a Karnaugh Map, we derive the fundamental Boolean equation governing the SR latch. This equation defines the next state mathematically:

$$Q_{n+1} = S + \overline{R} \cdot Q_n \quad (2.1)$$

Constraint: $S \cdot R = 0$ (They must never be 1 simultaneously, preventing the forbidden state).

Chapter 3

Gating and The Clock Signal

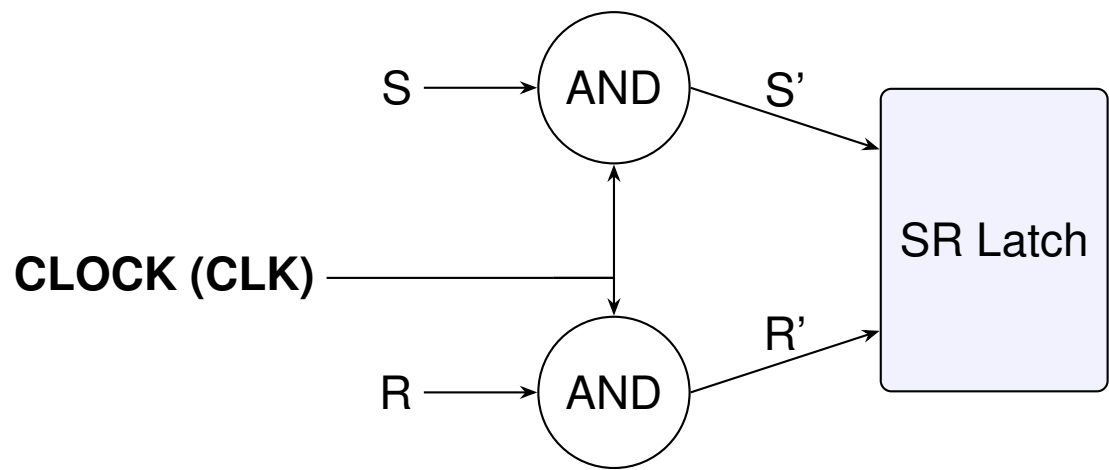
3.1 The Problem with Asynchronous Latches

The basic SR Latch we just designed is *asynchronous* and *transparent*. This means that the moment the S or R input changes, the output Q changes instantly (delayed only by the nanosecond propagation delay of the logic gates).

In a complex computer processor containing millions of memory elements, if every latch updated instantly whenever its inputs fluctuated, chaos would ensue. Electronic signals take different amounts of time to travel across a motherboard. We need a way to synchronize the chaos. We need a way to tell the memory elements: **"Do not look at your inputs until I tell you to."**

3.2 The Gated SR Latch

To solve the transparency problem, we add two AND gates to the front of the SR Latch, and tie one leg of each AND gate to a master control wire called the **Enable (EN)** or **Clock (CLK)**.



3.3 Level-Triggering Mechanics

The Gated SR Latch operates on a principle called **Level-Triggering**.

- **When CLK = 0 (LOW):** The AND gates output 0 regardless of what the external S and R signals are doing. The internal latch receives $S'=0$, $R'=0$ (The Hold State). The latch is deaf to the outside world. The data is locked and secure.
- **When CLK = 1 (HIGH):** The AND gates act as open windows. The external signals S and R pass straight through to the internal latch. The latch becomes *transparent* and updates its state to match the inputs.

3.3.1 The Fatal Flaw of Level-Triggering

Level-triggering solved the problem of random updates, but it introduced a new, deadly issue. A clock pulse stays HIGH (CLK=1) for a specific physical duration (e.g., 500 nanoseconds).

If the outputs of the latch are fed back into its own inputs through external combinational logic, the latch might update its state, feed that state back to itself, and update *again* multiple times before the 500ns clock pulse finally goes LOW.

This causes the output to oscillate wildly during the HIGH phase of the clock. This catastrophic failure is known as the **Race-Around Condition**.

Chapter 4

Edge-Triggering and Flip-Flops

4.1 The Evolution from Latch to Flip-Flop

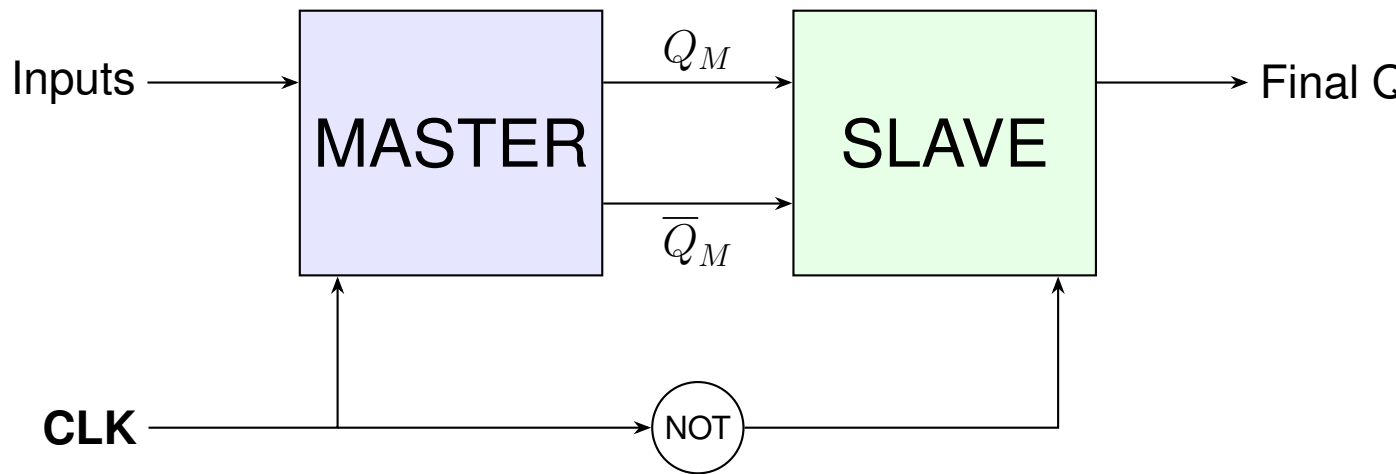
To completely eliminate the Race-Around condition, we must mathematically shrink the "window of transparency" from a long duration (Level-Triggering) to an infinitely small instant in time.

We do this by designing circuits that only respond to the exact moment the clock transitions from 0 to 1 (**Positive Edge-Triggered**) or from 1 to 0 (**Negative Edge-Triggered**).

Formal Definition: A memory element that is Edge-Triggered is formally classified as a **Flip-Flop**. A memory element that is Level-Triggered is a **Latch**.

4.2 The Master-Slave Architecture

The most robust method to create edge-triggering is the Master-Slave configuration. It involves placing two Gated Latches in series and inverting the clock signal between them.



4.3 Master-Slave Execution Physics

Let us rigorously analyze a **Negative-Edge Triggered** Master-Slave JK Flip-Flop to understand how it defeats the race-around condition.

4.3.1 Phase 1: The Clock is HIGH (CLK = 1)

- The Master latch receives CLK = 1. It becomes completely transparent and actively monitors the external J and K inputs. It updates its internal temporary states (Q_M).
- The Slave latch receives the inverted clock (CLK' = 0). It is completely deaf and locked in its Hold state.
- Because the Slave is locked, the final output Q **does not change**. The external feedback loop is physically broken. The Race-Around condition cannot occur because the changes in the Master never reach the outside world.

4.3.2 Phase 2: The Falling Edge (CLK transitions 1 → 0)

- At the exact nanosecond the clock falls, the Master latch receives CLK = 0. It instantly snaps shut, locking the inputs it just read into its memory.
- Simultaneously, the Slave latch receives CLK' = 1. It becomes transparent, reads the locked data from the Master (Q_M), and pushes it to the final output Q.
- The final state of the flip-flop updates **exactly once** per clock cycle, precisely on the falling edge.

Chapter 5

The Four Core Flip-Flops

5.1 1. The SR Flip-Flop (Set-Reset)

As derived previously, the SR flip-flop sets Q to 1 on S=1, resets Q to 0 on R=1, and holds on 00. The 11 state remains fundamentally forbidden. It is the building block but is rarely used in complex ICs due to the invalid state.

5.2 2. The JK Flip-Flop (The Universal Element)

The JK Flip-Flop was engineered specifically to solve the SR forbidden state anomaly. By routing the final Q and $\overline{Q}$ outputs back into the input gating structure, the flip-flop gains a revolutionary new capability.

When $J = 1$ and $K = 1$, the flip-flop enters **Toggle Mode**. Upon every active clock edge, the output logically inverts ($Q_{n+1} = \overline{Q}_n$). It alternates between 0 and 1, creating a perfect divide-by-two frequency circuit. This is the heart of all digital counters.

5.2.1 JK Characteristic Truth Table

Characteristic Equation: $Q_{n+1} = J\overline{Q}_n + \overline{K}Q_n$

J	K	Q_{n+1}	Operation
0	0	Q_n	Hold
0	1	0	Reset
1	0	1	Set
1	1	$\overline{Q_n}$	Toggle

5.3 3. The D Flip-Flop (Data / Delay)

The D Flip-Flop is the most widely used memory element in modern processors. It is used to build CPU Registers, Shift Registers, and Static RAM. It has only one data input (D).

It is constructed by taking an SR flip-flop and wiring an inverter between the S and R pins, such that $S = D$ and $R = \overline{D}$. This guarantees that the forbidden 11 state can never physically occur.

5.3.1 D Characteristic Truth Table

D	Q_{n+1}	Operation
0	0	Stores a 0
1	1	Stores a 1

Characteristic Equation: $Q_{n+1} = D$

Behavior: The output simply tracks the input, delayed by one clock pulse. It acts as a perfect data buffer.

5.4 4. The T Flip-Flop (Toggle)

The T Flip-Flop is used heavily in Binary Counter circuits. It is constructed by tying the J and K pins of a JK flip-flop together into a single master pin (T).

5.4.1 T Characteristic Truth Table

Characteristic Equation: $Q_{n+1} = T \oplus Q_n$

T	Q_{n+1}	Operation
0	Q_n	Hold State
1	$\overline{Q_n}$	Toggles State

Chapter 6

The Master Synthesis Tool

6.1 Understanding Excitation Tables

While Truth Tables predict the *future* based on the present, **Excitation Tables** dictate the *present* required to achieve a specific future.

If you are an engineer designing a Vending Machine, and the machine is currently in State 0 and needs to move to State 1 when a coin is inserted, you must consult the Excitation Table. It tells you exactly what electrical voltages you must apply to the Flip-Flop pins to force that exact transition.

6.2 The Master Excitation Matrix

You must memorize this matrix to perform Phase II hardware conversions and Phase VI Finite State Machine (FSM) synthesis. It is the absolute core of Sequential Systems.

***** END OF PHASE I MASTER
MANUAL *****

Transition		SR FF		JK FF		D FF	T FF	
Q_n	Q_{n+1}	S	R	J	K	D	T	
0	0	0	X	0	X	0	0	
0	1	1	0	1	X	1	1	
1	0	0	1	X	1	0	1	
1	1	X	0	X	0	1	0	